

1 Part 1: Bright Solitons in the Non-Linear Schrodinger Equation

Here we explore numerical simulations for interacting wave solutions via the non-linear Schrodinger equation. In 1 spatial dimension, we use,

$$\frac{\partial \psi}{\partial t} = i \left(\frac{\partial^2 \psi}{\partial x^2} - g |\psi|^2 \psi \right), \quad (1)$$

where $\psi(x, t)$ is the complex wave function and g is the interaction strength. We work here with attractive interactions which is where $g \leq 0$. This gives us "bright soliton" solutions which are stable wave packets. In this section, we explore their dynamics under varying conditions and with respect to 3 key quantities, interaction strength, velocity and relative phase.

1.1 A: Evolution Implementation

We aim to solve equation 1 on the domain $-L/2$ to $L/2$ where we set $L = 20$ and with Dirichlet boundary conditions $\psi(-L/2, t) = \psi(L/2, t) = 0$. We use RK4 as with assignment 1 and approximate the spatial derivative via the central difference method. We also set a grid size of $N = 128$, a time step of 0.01 and a max time of 20. We save data every 20 steps.

To implement evolution I used 2 separate functions. Firstly we define the calculateRHS function which finds the right hand side of equation 1 via the derivative term and the nonlinear term and combines them after multiplying by i . We also set our boundary conditions to 0 for the RHS as the time derivative of those points must be 0. Next we use the RungeKutta4 method to update the complex vector ψ using the RHS function to find the time derivative at the 4 times required.

1.2 B: Plane Wave

To solve for the evolution of a plane wave we must use the aforementioned functions alongside an initialisation function. We initialise the wavefunction as $\psi(x, 0) = \cos(\pi x/L)$. Note that we also normalise this after initialising to ensure it remains a physical quantum system.

We then evolve it via RK4 and calculate the wave amplitude. We loop over 5 values of g (-10, -3, -2, -1, -0.5, 0) which show a large range of interaction strengths. The evolutions are saved in txt files and then animated and saved as a gif in the Plots folder. We name it nlse_animation_plane_wave.gif This shows each of the plane waves with different g values as they evolve. We can observe that they all remain stationary as expected but warp by becoming more localised at $x = 0$ and then spreading out periodically.

This is because under strong attraction $|g| \gg 1$ the system begins forming a bright soliton-like structure which resembles a localized blob of high amplitude around $x = 0$ and then spreads back out again.

More physically, the system prefers to minimise energy, and a strong attractive interaction lowers energy by clumping the wavefunction together at the centre of the plane wave. This results in a localized bound state — the hallmark of bright solitons. And noticeably, the stronger the interaction, the tighter the localization. When g becomes very large we see that from the SE, the dynamics are almost entirely determined by the attractive term, which means the wavefunction becomes attracted to a single point.

For future questions we set $g = -1$.

1.3 C: Wave Packet

Next we investigate the dynamics of a wavepacket which is defined by the initial condition,

$$\psi(x, 0) = \sqrt{2} e^{iux} \text{sech}(x). \quad (2)$$

We plot the dynamics as before as an animation for 5 values of u (-1, -0.5, 0, 0.5, 1) to give us both positive, negative and a 0 value. This is saved in the plots folder accordingly. We can clearly observe that for $u = 0$, the wavepacket stays stationary while for negative values it moves to the left and for positive values it moves to the right. This clearly indicates that u represents the velocity of the wavepacket. This is hence our bright soliton.

Additionally, we can observe that due to our Dirichlet boundary conditions, the soliton 'bounces' off the walls when it gets close. We can hence track the peak position of the wave amplitude over time for our values of u and observe the effects. We plot the resulting figure in 1. This clearly demonstrates the effect of u is to add a velocity to the wave packet.

For the next question we set $u = 0.1$.

1.4 D: Solitons

Now we expand our simulation to solve for the evolution of 2 solitons. To do this we simply change our initial condition to,

$$\psi(x, 0) = \sqrt{2} \exp(iux) \text{sech}(x + L/4) + \sqrt{2} \exp(-iux + i\theta) \text{sech}(x - L/4). \quad (3)$$

This equation represents 2 solitons beginning halfway on either side of the lattice, moving towards each other with velocity u and with a relative phase of θ . Again we explore the dynamics of the solution by animating their motion for 5 values of

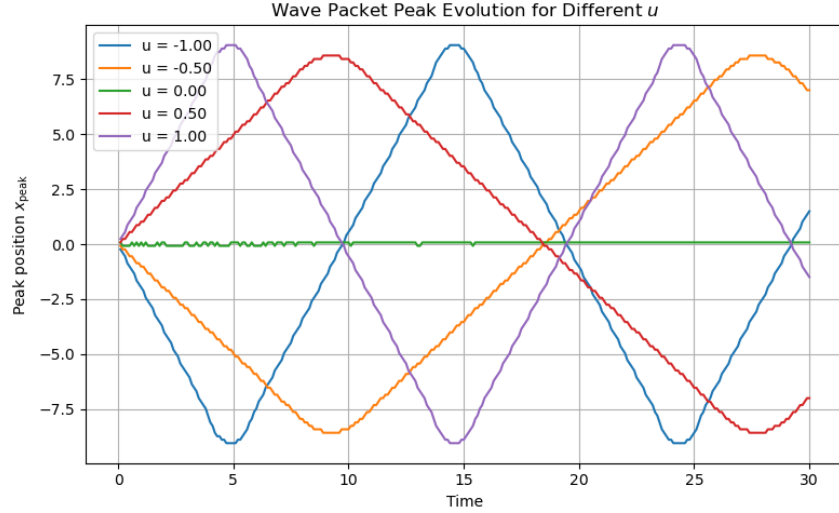


Figure 1: Plot of the peak position of the wavefunction over time for the wave packet initial condition. We can clearly observe that for $u = 0$, the packet does not move from the centre of the grid. Whereas for large u , the peak moves faster and bounces off the walls earlier. We also see symmetry around $u = 0$ indicating that the sign of u represents the packets velocity.

theta ($2 * \pi, \pi, 2.0 * \pi/3.0, \pi/2.0, 0$). We can clearly observe from the animation that for theta close to 0 and 2π the solitons merge while for others the solitons bounce, with the largest bounce occurring at $\theta = \pi$ which is when the solitons are most out of phase. To investigate this further we can monitor the positions of the 2 peaks and the distance between them in figures 2 and 3.

This demonstrates what we observe qualitatively from the animation, where the solitons get closer and bounce/merge. We can also plot the distance of closest approach of the solitons against our values of θ to determine what their behaviour looks like and the dependence on θ . The results are presented in figure 4. Here we see the same behaviour as when the solitons are in phase, they get closer than when they are out of phase.

Using these observations we can hypothesise why the solitons appear to merge or bounce for different phases. It is clear that the relative phase influences how the solitons interfere when they overlap and we can observe that for $0, 2\pi$ they constructively interfere and hence 'merge' while for other values they destructively interfere (or a mix) and hence a single peak is not evident, instead bouncing off each other. Due to the nonlinearity in the NLSE, the constructive interference causes them to temporarily form a bound state, appearing to "merge" before separating again.

We can think of the soliton interaction as governed by a phase-dependent effective potential which is attractive for in phase and repulsive for out of phase. This is because the nonlinear interference term in the NLSE can be viewed as a coherence-mediated force — constructive coherence increases local intensity and tightens interaction, while destructive coherence disperses it. This is the interaction that causes the appearance of bouncing or merging.

This is also clear from the form of the initial wavefunction where the primary difference in the exponential term is the relative phase. Clearly $e^{i2\pi} = 1$ and so when θ is a multiple of 2π , the terms become equivalent and can combine/merge. Alternatively, when θ is an odd multiple of π , we get a destructive term via $e^{i\pi} = -1$, which causes them to bounce.

We can also understand this interaction more formally using the Hamiltonian of the NLSE from lecture notes:

$$H[\psi] = \int \left(\frac{1}{2} |\partial_x \psi|^2 - \frac{g}{2} |\psi|^4 \right) dx$$

The nonlinear interaction term $-\frac{g}{2} |\psi|^4$ implies that constructive interference (as occurs for in-phase solitons) leads to a stronger, more localized wave profile, thereby lowering the interaction energy. This corresponds to an effective attraction, and manifests as the appearance of merging or temporary bound states. In contrast, destructive interference (e.g., for a relative phase of π) results in a reduced local intensity and smaller change in interaction energy, leading to effective repulsion and bouncing dynamics.

We can try to show this mathematically where if the solitons are in phase we have at time t_0 , where they are both centred on 0,

$$\psi(x, t_0) = \sqrt{2}e^{iux} \operatorname{sech}(x) + \sqrt{2}e^{-iux} \operatorname{sech}(x) = 2\sqrt{2} \cos(ux) \operatorname{sech}(x),$$

which represents constructive interference at $x = 0$ while if they are out of phase we have,

$$\psi(x, t_0) = \sqrt{2}e^{iux} \operatorname{sech}(x) - \sqrt{2}e^{-iux} \operatorname{sech}(x) = 2\sqrt{2}i \sin(ux) \operatorname{sech}(x),$$

which represents destructive interference at $x = 0$ which is why they merge in the first case and bounce in the 2nd case.

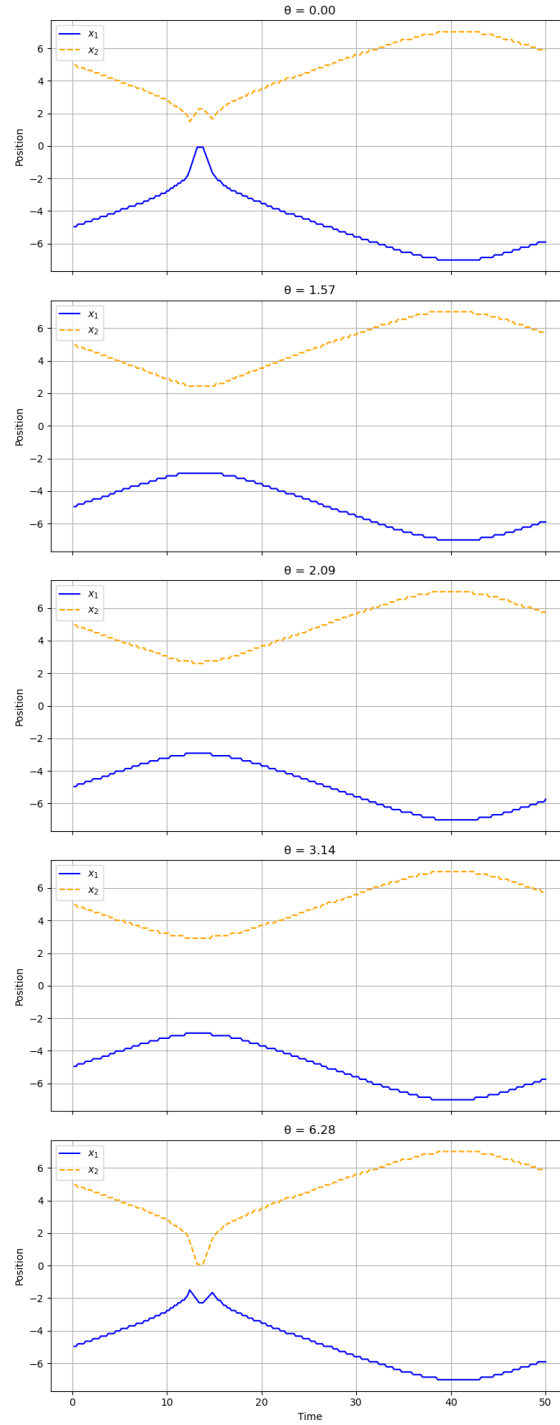


Figure 2: Plot of the peaks of the solitons over time for various relative phases, we can observe for $\theta = 0, 2\pi$ we have the closest peaks but we do not see evidence of the merge. This is likely an artifact from the method which we use to calculate the peaks. Regardless, we see the behaviour of 2 solitons moving towards each other and interacting differently depending on θ .

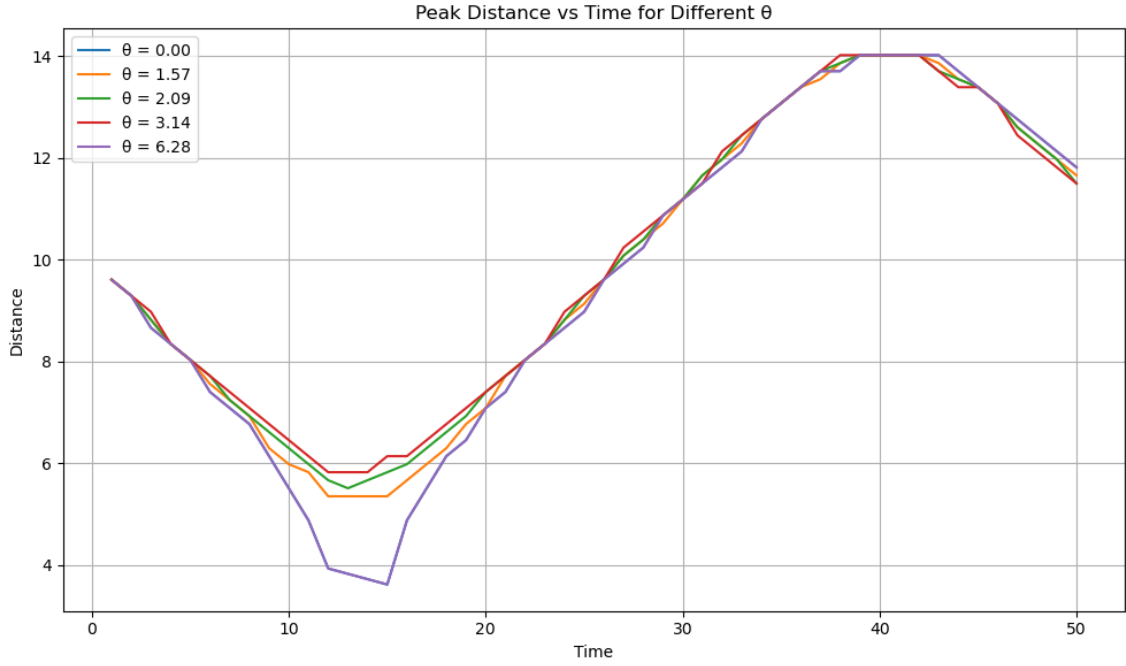


Figure 3: Plot of the distance between peaks vs time to demonstrate how the solitons interact and how close they get before merging or bouncing. We can clearly see the solitons moving towards each other before bouncing for off phase cases and then getting closer together for 0 and 2π . Note the lines for 0 and 2π are directly overlapping.

1.5 Conclusion

To conclude, we numerically investigated the dynamics of bright solitons in the one-dimensional non-linear Schrödinger equation with attractive interactions. Starting from simple plane wave and wave packet initial conditions, we validated the emergence of stable, localized structures characteristic of bright solitons. By varying interaction strength, initial velocity, and relative phase between solitons, we demonstrated how each parameter affects the evolution and interaction of these nonlinear wave packets.

We observed how the interaction strength causes localisation of the solution and peaking. We also showed how altering u affects the velocity of the Soliton and the interaction of solitons with our boundary conditions (bouncing off with some distortion). Most notably, we observed phase-dependent soliton interactions, where solitons merge or repel based on their relative phase—revealing an effective coherence-mediated interaction potential. These findings highlight the rich dynamical behaviour that arises from the interplay between dispersion and nonlinearity, and the crucial role of phase in governing soliton interactions.

Finally, all code is available on [Github](#) and the animations are found in the plots folder. We omit them in this report for brevity and space limitations.

2 Part 2: The Transverse Ising Spin Model

Our next task is to investigate the transverse Ising spin model in 1 dimension to observe a quantum phase transition. Traditionally when we work with the standard Ising model, thermal phase transitions arise when symmetric properties of the equilibrium state change. After a critical point, the system becomes disordered, manifesting as a point of non-analyticity in the free energy as a function of temperature. In our case (many body quantum systems), an analogous effect occurs at 0 temperature. We can observe a point of non-analyticity in the ground state energy associated with a change in the symmetric properties of the ground state wave function's energy.

Note here that a sufficient condition for non-analyticity is that it has a point where the derivative is not defined. The 1D transverse Ising model is what we use to explore this, given by,

$$\hat{H} = -\omega \sum_{m=0}^{N-1} \hat{\sigma}_z^m - g \sum_{m=0}^{N-1} \hat{\sigma}_x^m \hat{\sigma}_x^{m+1}, \quad (4)$$

where,

$$\hat{\sigma}_z^{(m)} = I^{\otimes m} \otimes \begin{pmatrix} 1/2 & 0 \\ 0 & -1/2 \end{pmatrix} \otimes I^{\otimes (N-1-m)} \quad \text{and} \quad \hat{\sigma}_x^{(m)} = I^{\otimes m} \otimes \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix} \otimes I^{\otimes (N-1-m)} \quad (5)$$

are the z, x pauli spin matrices respectively for a single spin half particle located at the m th lattice site. I is the 2×2 identity, and $\otimes$ represents the tensor product. We impose periodic boundary conditions so our matrices ‘loop’ as,

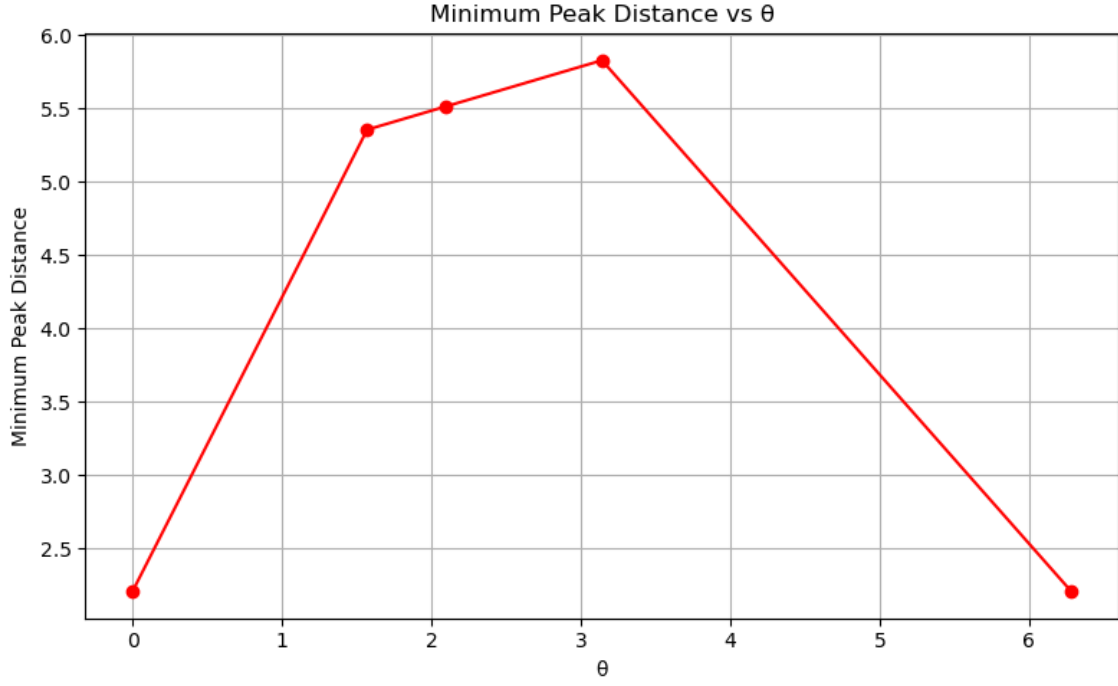


Figure 4: Minimum peak distance of the solitons for our 5 values of θ . We can clearly see when the relative phase is in phase, the solitons get closer than when the phase is out of phase. The dynamics are also symmetric around π as expected.

$$\hat{\sigma}_x^N = \hat{\sigma}_x^0. \quad (6)$$

The first term in the Hamiltonian describes the coupling of 1 atom to an external field that drives a transition between 2 single-particle states. The 2nd term describes the nearest-neighbour spin interaction with $g \geq 0$. This is analogous to the standard Ising model with a magnetic field and nearest neighbour interactions. We set $\omega = 1$.

We wish to explore the computational scaling of creating the Hamiltonian matrix as a function of N , how the ground state energy depends on g and hence if we observe a point of non-analyticity and finally how the dynamics of a quantum state depend on initial g and if we quench (rapidly change) g so the state is not longer in the ground state.

2.1 A: Execution Time

Our first task is to construct the Hamiltonian H . We begin by defining a Matrix class from prior assignments. We do not modify the data type as our matrices also still use real values, it is only our state vectors which will be complex. We add methods to extract elements from the matrix, print out the matrix and add matrices as standard operations. We also reuse our eigendecomposition code to find the eigenvalues and vectors using LAPACK. The primary novel addition for this matrix class is the inclusion of a tensor product method. This is a method we can call on a matrix from this class to tensor product it with another matrix (from the same class). To do this, we create a new matrix of the appropriate size (rows1 times rows2 and same for columns) and then iterate through a for loop where we find each element via elementwise multiplication. This is all found in the Matrix.hpp file.

Our next step is to find each term of the Hamiltonian, to do this we find the σ_z^m matrices and the $\sigma_x^m \sigma_x^{m+1}$ matrices. For the former, we iterate from 0 to $N - 1$ starting with a 1 and then tensor product the matrix with the Identity or Z depending on the index. Similarly, for the latter, we use the hint that,

$$\hat{\sigma}_x^{(m)} \hat{\sigma}_x^{(m+1)} = I^{\otimes m} \otimes \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix} \otimes \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix} \otimes I^{\otimes (N-2-m)}. \quad (7)$$

to iterate from 0 to $N - 1$ and times by I unless the index equals m or $m + 1$ with appropriate boundary conditions. We do so similarly for x and z matrices. Then to find the first term, we use a for loop over N to find each matrix for m and add them together, noting that $\omega = 1$. For the 2nd term, we do the same process with $m, m + 1$ and multiplying by g . These processes are found in part2_functions.hpp.

We then add these terms together and find the Hamiltonian, from which we can find the eigenvalues and eigenvectors. We time this process as we loop over values of N and plot our results in figure 5. We can clearly observe an exponential trend which is logical as the size of our matrix scales as 2^{2N} . We also observe the time taken for the first loop to initialise which often adds to the time taken for the first iteration. A similar trend was observed in the Ising model when we used OpenMP for multithreading.

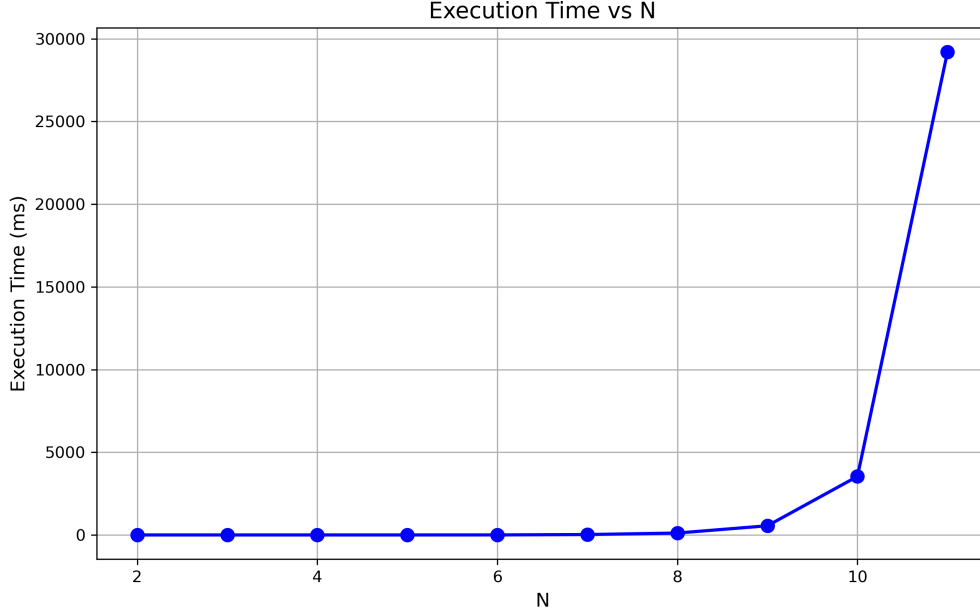


Figure 5: Plot of the execution time of constructing the Hamiltonian demonstrating an exponential time taken with a small overhead for the first iteration. This suggests the construction of the Hamiltonian scales logically indicating that the tensor product operation is likely functioning correctly.

2.2 B: Ground State Energy

Our next goal is to observe the quantum phase transition. We do this by finding the ground state energy, E_0 , and plotting E_0/N as well as $\frac{\partial^2 E_0}{\partial g^2}/N$ as function of g for $0 < g < 8$. We also wish to plot against the analytical result in the thermodynamic limit,

$$\lim_{N \rightarrow \infty} \left\{ \frac{\varepsilon_0}{N} \right\} = -\frac{1}{2\pi} (2+g) E \left[\frac{8g}{(2+g)^2} \right], \quad (8)$$

where

$$E[x] = \int_0^{\pi/2} d\theta \sqrt{1 - x \sin^2(\theta)}. \quad (9)$$

We find the second derivative of a function via the central difference method as used in prior work. To find the ground state energy, we find the Hamiltonian for different g and then take the first eigenvalue. We then store this as we loop over g . To find the elliptical integral, we use a numerical approximation to the integral via simpsons rule for accuracy. Our results for the energy are shown in figure 6(a).

We clearly see an exact match for all g , indicating that the construction of the Hamiltonian is very accurate. We can see a sharp change in concavity around $g = 2$. This points to a location of non-analyticity which we investigate further via the 2nd derivative. This is shown in figure 6(b).

We clearly observe a point of non-analyticity in the theoretical energy values and a corresponding significant dip in the numerical values. We also see an almost exact match for most values of g except around the critical point. The discrepancy is likely because we are not at the thermodynamic limit where $N \rightarrow \infty$.

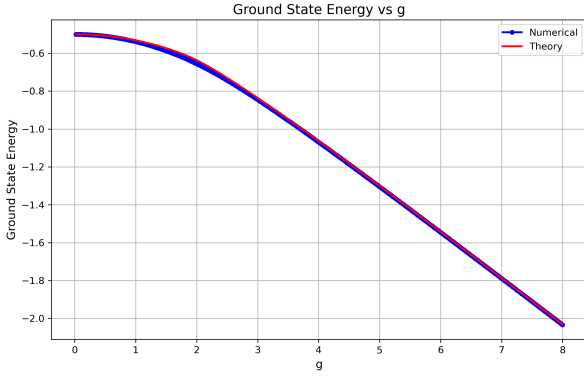
These results are clearly indicative of the fact we are observing a phase transition dependent on g . We can clearly see that the transition happens at $g = 2$, which is when $g = 2\omega$. This follows from our definition the Spin matrices as if we were to instead use pauli matrices with 1 in each element, the phase transition occurs at $g = \omega$. This demonstrates that when the interaction strength becomes stronger than the individual atom coupling, we observe a phase transition. This indicates that when $g > 2$, we enter a paramagnetic type regime, where neighbouring spins wish to aligned together in the x direction. While for $g < 2$ we are in a ferromagnetic regime where neighbouring spins prefer to be paralleled aligned in the z direction.

We can use the Hamiltonians in each of the different representations to interpret what the phase transitions represent.

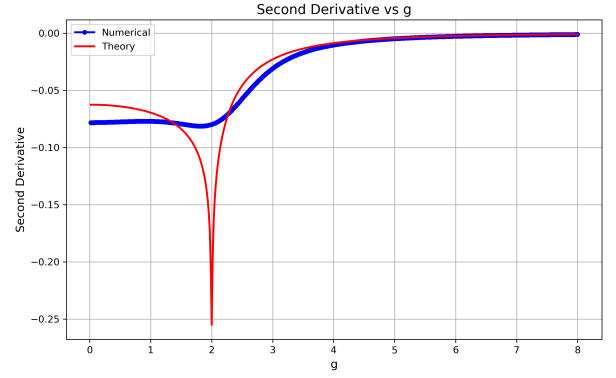
In the spin representation our Hamiltonian is,

$$\hat{H} = - \sum_{m=1}^L [J (\hat{\sigma}_+^m \hat{\sigma}_-^{m+1} + \hat{\sigma}_-^m \hat{\sigma}_+^{m+1}) + J\gamma (\hat{\sigma}_+^m \hat{\sigma}_+^{m+1} + \hat{\sigma}_-^m \hat{\sigma}_-^{m+1}) + 2\omega \hat{\sigma}_+^m \hat{\sigma}_-^m]$$

while in the fermionic representation our Hamiltonian is,



((a)) Plot of the ground state energy against g and with the theoretical values at the thermodynamic limit. We see an almost exact match and a noticeable change in concavity around $g = 2$, indicative of a point of non-analyticity. We also see a slight discrepancy around this point due to the fact we are not at the thermodynamic limit.



((b)) 2nd derivative of the ground state energy as a function of g with the theoretical and numerical data plotted together. We see evidence of the point of non-analyticity in the theoretical equation and a corresponding dip in the numerical simulations. This is indicative of a phase transition. Slight discrepancies are caused as we are not in the thermodynamic limit.

Figure 6: Ground state energy and its second derivative as functions of g .

$$\hat{H} = - \sum_{m=1}^L \left[J \left(\hat{c}_m^\dagger \hat{c}_{m+1} + \hat{c}_{m+1}^\dagger \hat{c}_m \right) + J\gamma \left(\hat{c}_m^\dagger \hat{c}_{m+1}^\dagger + \hat{c}_{m+1} \hat{c}_m \right) + 2\omega \hat{c}_m^\dagger \hat{c}_m \right]$$

and in momentum space we have

$$\hat{H} = -2 \sum_{\substack{k \in K, \\ k > 0}} \left[(\omega + J \cos k) \left(\hat{c}_k^\dagger \hat{c}_k + \hat{c}_{-k}^\dagger \hat{c}_{-k} \right) + iJ \sin k \left(\hat{c}_k^\dagger \hat{c}_{-k}^\dagger + \hat{c}_{-k} \hat{c}_k \right) \right].$$

Note that these equations are from the lecture notes directly (where J represents the interaction strength rather than g and we set $\gamma = 1$ for the transverse spin model). We can begin with the spin representation where we have terms for neighbouring spins $m, m+1$ and a term for each individual spin site m . The strength of the interactions between m and $m+1$ is mediated by the value of J while the strength of single spin preferred alignment is mediated by ω . Hence it is clear from the form of the Hamiltonian that there are 2 distinct phases that must occur. Firstly when $J < \omega$, the interaction strength which prefers opposite alignment of spins is weaker than the strength of the external field which prefers a spin direction. This gives us a “quantum ordered” system where the spins are aligned parallel. On the other hand when $J > \omega$, we have a stronger interaction strength than the external field. However, this form of the Hamiltonian does not present us with a preferred spin direction as we have 4 terms representing opposite and same alignment for neighbouring spins. These terms all equally contribute as we have set $\gamma = 1$. These are our 2 distinct phases in this system in terms of the spin operator Hamiltonian. In this case when $J = \omega$, we are in a gapless regime where the phase is at a critical point.

As the physical system does not change between representations, we would expect a similar interpretation of the phases in the other representations. With the fermionic representation, spins are mapped to whether fermions occupy the site or not. This hence becomes analogous to a Bose-Einstein condensate type problem, where phases now represent superconducting phases. This is because the whole spin system is always interpreted at 0 temperature and so when considering fermions we must consider superconducting properties. When we are now in the “paramagnetic” regime ($J > \omega$), the fermions are in a trivial superconducting phase which means there are no edge states. whereas in the ferromagnetic regime ($J < \omega$) we can consider the model to be a topological superconducting phase with Majorana edge modes in open chains. This is the Kitaev chain model of a 1D p-wave superconductor. In this model the typical names for the phases are the strong coupling and weak coupling phases for strong and weak interaction strengths respectively. More colloquially, one can interpret the fermionic representation phases as either occupation in sites being the same in neighbouring sites (groups of fermions together) when $J \ll \omega$.

We can consider the Hamiltonian in 2 different regimes, when ω is negligible and when J is negligible. We see that in the first case we have

$$\hat{H} = - \sum_{m=1}^L \left[J \left(\hat{c}_m^\dagger \hat{c}_{m+1} + \hat{c}_{m+1}^\dagger \hat{c}_m + \hat{c}_m^\dagger \hat{c}_{m+1}^\dagger + \hat{c}_{m+1} \hat{c}_m \right) \right]$$

and in the latter we have,

$$\hat{H} = - \sum_{m=1}^L 2\omega \hat{c}_m^\dagger \hat{c}_m.$$

The latter clearly demonstrates that the Hamiltonian pushes the fermionic sites to all be occupied, which gives us the quantum ordered system. The first Hamiltonian gives us terms which only contribute if the fermions in neighbouring sites are both opposites or both the same. As such, we cannot directly tell from this Hamiltonian what the preference is and instead consider the prior argument from an empirical standpoint and from the XY Hamiltonian.

Finally, when we consider momentum space, we use k -space to denote which regime we are in, and we change our perspective from spin alignment or fermionic occupation to instead vibrations and wave number occupation (i.e. how much momentum the waves have in the “lattice”). This is a far more abstract interpretation of the system but physically must still give rise to the same phase transitions. In this interpretation, the ordered system would indicate modes where many similar wave numbers k are “occupied” in momentum space, while the disordered system would indicate a spread over a large variety of k wave numbers. From the Hamiltonian, we see that the J mediates the kinetic and imaginary terms while ω mediates the potential term. As such, in momentum space, the dispersion relation reflects the competition between these two energy scales. When $J > \omega$, the system favors alignment along the x -direction (ferromagnetic order in the original spin picture), leading to well-defined k -modes with minimal dispersion — a signature of coherence and order. Conversely, when $\omega < J$, the transverse field dominates, favouring spin flips and delocalized excitations across k -space, which corresponds to a disordered, paramagnetic phase. The critical point between these regimes is marked by a gap closing at specific k -values (e.g., $k = 0$ or $k = \pi$), where long-wavelength modes become massless, signalling the onset of a quantum phase transition.

Again we can consider what the Hamiltonian looks like under the 2 regimes for $\omega \gg J$ and $J \gg \omega$,

$$\hat{H} = -2 \sum_{\substack{k \in K, \\ k > 0}} \omega \left(\hat{c}_k^\dagger \hat{c}_k + \hat{c}_{-k}^\dagger \hat{c}_{-k} \right),$$

and

$$\hat{H} = -2J \sum_{\substack{k \in K, \\ k > 0}} \left[\cos k \left(\hat{c}_k^\dagger \hat{c}_k + \hat{c}_{-k}^\dagger \hat{c}_{-k} \right) + i \sin k \left(\hat{c}_k^\dagger \hat{c}_{-k} + \hat{c}_{-k}^\dagger \hat{c}_k \right) \right].$$

The former clearly wants each value of k and $-k$ to be occupied for all k . However the 2nd regime is unclear what is happening as we have the non-diagonal term of $-k$ and k together. We can hence look at the energy terms of the Hamiltonian in terms of ω, J and k . We can set $J = 1$ and then look at what happens to the energies given by

$$\varepsilon_k = \pm 2\sqrt{(\omega + J \cos k)^2 + (J \sin k)^2}$$

and plot the regimes. We see this in figure 7. This does not help us much as it just demonstrates that the system might prefer for energies across all k to be occupied. The reason this is ok is because the eigenstates are not simple in terms of k and instead non diagonal with terms that include both k and $-k$ which becomes difficult to interpret physically. In particular, the eigenstates are given by,

$$\hat{\gamma}_k = u_k \hat{c}_k + v_k \hat{c}_{-k}^\dagger, \quad \hat{\gamma}_k^\dagger = u_k^* \hat{c}_k^\dagger + v_k^* \hat{c}_{-k} \quad (10)$$

with

$$u_k = \cos \theta_k, \quad v_k = i \sin \theta_k \quad (11)$$

and angle θ_k defined by:

$$\tan(2\theta_k) = \frac{J \sin k}{\omega + J \cos k}. \quad (12)$$

Which is clearly quite complicated and with non diagonal terms which make it difficult to interpret. As such, we are unable to give a satisfying interpretation of the phase transitions in momentum space and instead leave interpretations to the XY Hamiltonian.

Note that some of this analysis comes from¹.

¹<https://arxiv.org/pdf/1012.0653>

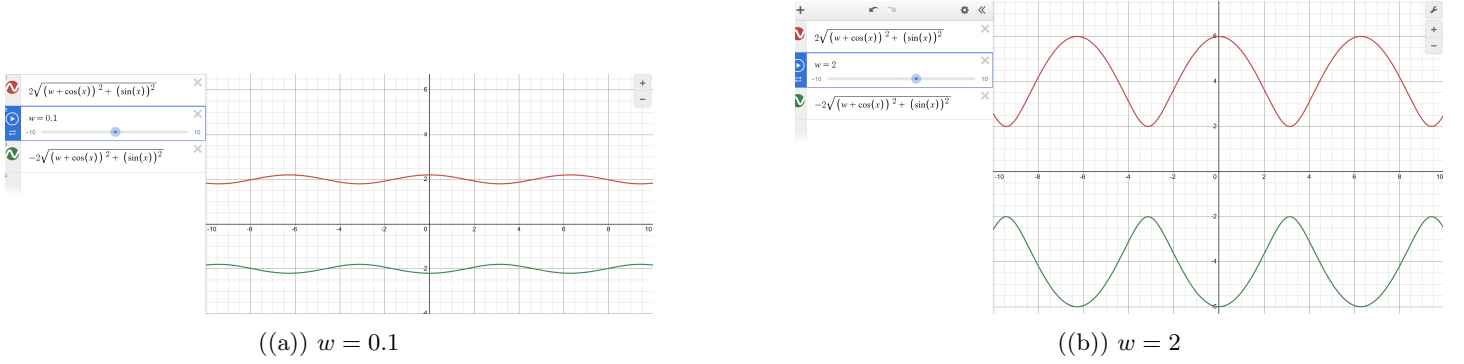


Figure 7: Comparison of energies for different values of w . We see that there are always negative energies and a symmetry around 0 which indicates that the system wants all values of k occupied at all times.

2.3 C: Evolution of Observables

Our final task is to use this Hamiltonian to investigate the evolution of wavefunctions. To explore the non-equilibrium dynamics of a system, a common method is to start in the ground state of a Hamiltonian and quench (rapidly change) a parameter so it is no longer in the ground state. In our case, we do so with g . We investigate 10 cases split into 2 sections. Firstly we start with $g = 0$ and then evolve according to $g = 0, 1, 2, 3, 4$.

Computationally, we first find the ground state wavefunction via the first eigenvector associated with the lowest eigenvalue. We then implement RK4 by first finding the derivative, via $-iH|\psi\rangle$ using matrix vector multiplication. We then use RK4 and update the state vector accordingly. To observe dynamics we aim to track the time evolution of 3 observables, namely,

$$S_z = \sum_{m=0}^{N-1} \langle \hat{\sigma}_z^{(m)} \rangle, \quad (13)$$

$$S_x = \sum_{m=0}^{N-1} \langle \hat{\sigma}_x^{(m)} \rangle, \quad (14)$$

and

$$C_{xx} = \sum_{m=0}^{N-1} \sum_{\substack{n=0 \\ n \neq m}}^{N-1} \langle \hat{\sigma}_x^{(m)} \hat{\sigma}_x^{(n)} \rangle. \quad (15)$$

To calculate these, we find the σ matrices as before and then find the expectation value via the matrix vector multiplication method and the dot product method and sum over all m, n . We run the C++ code over 2 for loops for the initial g value and the evolution g values to produce the plots given in figures 8(a) and 8(b). We note that the results are sensitive based on the time steps chosen, the lattice size and other hyperparameters relevant to this problem. Specific choices for our results are shown in code and easily modifiable if required.

The salient features of these observables are firstly that S_x remains 0 in all cases regardless of initial g or evolution g . Additionally, we observe consistent fluctuations and oscillations in the non-zero observables, with greater amplitude appearing when the evolution g is further from the initial g . This is logical as a stronger 'quench' would introduce stronger oscillatory dynamics of observables. These oscillations occur as a result of Rabi oscillations where a change to the Hamiltonian causes the system to oscillate between eigenstates of the new Hamiltonian.

Of particular interest is why — based on the Hamiltonian — we observe such dynamics. Firstly, we can see that the Hamiltonian is invariant under a global x flip (if we change all $\sigma_x \rightarrow -\sigma_x$).

As such, if the initial state respects this symmetry — which in our case it does as we use the ground state which is equivalent to the thermal state at equilibrium — then expectation values of values of operators which odd under this symmetry must vanish. This means the expectation value of σ_x must be 0 and hence S_x must be 0.

We can prove this mathematically. Let's denote the symmetry operator as $\hat{U}$, which performs the global x flip:

$$\hat{U} \hat{\sigma}_x^m \hat{U}^\dagger = -\hat{\sigma}_x^m, \quad \hat{U} \hat{\sigma}_z^m \hat{U}^\dagger = \hat{\sigma}_z^m.$$

Since $\hat{H}$ commutes with $\hat{U}$, we have:

$$[\hat{H}, \hat{U}] = 0.$$

So if $|\psi\rangle$ is a sum of eigenstates of $\hat{H}$, then $\hat{U}|\psi\rangle = |\psi\rangle$, and the expectation value transforms as:

$$\begin{aligned} \langle \hat{\sigma}_x^m \rangle &= \langle \psi | \hat{\sigma}_x^m | \psi \rangle = \langle \psi | \hat{U} \hat{\sigma}_x^m \hat{U}^\dagger | \psi \rangle \\ &= \langle \psi | (-\hat{\sigma}_x^m) | \psi \rangle = -\langle \hat{\sigma}_x^m \rangle. \end{aligned}$$

and hence $\langle \hat{\sigma}_x^m \rangle = 0$.

On the other hand, we clearly observe non-zero dynamics for C_{xx} . This is logical as if the effect of a global spin flip on a single x spin is odd, then the impact of a global spin flip on neighbouring x spins is even (i.e. does change the Hamiltonian). Therefore, even if the symmetry enforces $\langle \sigma_x^m \rangle = 0$, it does not forbid correlations between distant spins in the x -direction which encode C_{xx} . These correlations are physical and observable, and in fact encode the onset of long-range order in the ferromagnetic phase (for large g). As such, the value for g has a large influence on the strength of these correlations and hence we observe non-zero time dynamics.

In a similar fashion, the Hamiltonian is clearly not invariant under a global z flip ($\sigma_z \rightarrow -\sigma_z$), which is why we observe oscillatory behaviour for $\langle \sigma_z^m \rangle$.

We can also argue this idea from a different perspective, we can see that the form of the Hamiltonian is,

$$\hat{H} = -\omega \sum_{m=0}^{N-1} \hat{\sigma}_z^m - g \sum_{m=0}^{N-1} \hat{\sigma}_x^m \hat{\sigma}_x^{m+1}, \quad (16)$$

which has 2 terms, firstly a term which pushes the system to be aligned in the z direction (which hence provides no contribution to expectation values of x). Secondly, it includes a term which prefers spins to be aligned in the x direction. This second term indicates to us what is happening in terms of the physical system. With just this term, the system would prefer a state of either $|\rightarrow \cdots \rightarrow\rangle$ or $|\leftarrow \cdots \leftarrow\rangle$ in the x direction with equal energy. This is because these states have the lowest energy by the Hamiltonian (if ω is negligible).

And from either of these states, the system should not make an x flip as it would only increase its energy. However, this contradicts the idea that C_{xx} should be non-zero and changing due to the symmetry of the Hamiltonian. As such, this logic may be flawed but it is what was discussed in tutorials.

This interpretation holds for when we begin with $g = 0$, as we would start with $|\uparrow \cdots \uparrow\rangle$ which is an equal superposition of rights and lefts and hence S_x should remain 0. However, when we begin with $g = 4$, the initial state would be in a superposition with contributions from right and left in the x direction, which should not change the C_{xx} quantity. This is clearly wrong as we do observe non-zero dynamics here, as such there must be something faulty with this interpretation in the case where $g \neq 0$ to begin.

2.4 Conclusion

In conclusion, this part of our task centred on the Transverse Ising Spin Model in 1 dimension, looking for a quantum phase transition. We observed this effect at $g = 2\omega$ due to our definition of the spin matrices. This follows logically from the form of the Hamiltonian and suggests the model has 2 distinct phases analagous to the ferromagnetic and paramagnetic phases of the standard Ising model. This transition was observed via a point of non-analyticity of the 2nd derivative of the ground state energy.

Our next task was to evolve the ground state wavefunction of a Hamiltonian once we quench the system. This allowed us to see the effect of g on observable dynamics. The primary motivation for such a task was to investigate symmetric properties of the Transverse spin Ising model, where we clearly observe S_x symmetry but not C_{xx} symmetry or S_z symmetry. This follows directly from the form of the Hamiltonian as it is invariant under a global x flip, implying that spin x operators are odd. We also observed how the strength of quenching influenced the amplitude of oscillations of observables.



Figure 8: Comparison of time dynamics of observables for different initial g values and varying evolution g values. We observe a 0 value for S_x for all g values. We then see for a larger difference between initial and evolution g , the oscillatory behaviour occurs with greater amplitude. This is logical as the strength of 'quenching' is the associated strength of dynamical behaviour. This information also shows us the symmetric properties of the transverse spin Ising model as described in the section above.